BPM2_NB_ESOFT QUIZ 1_S2026

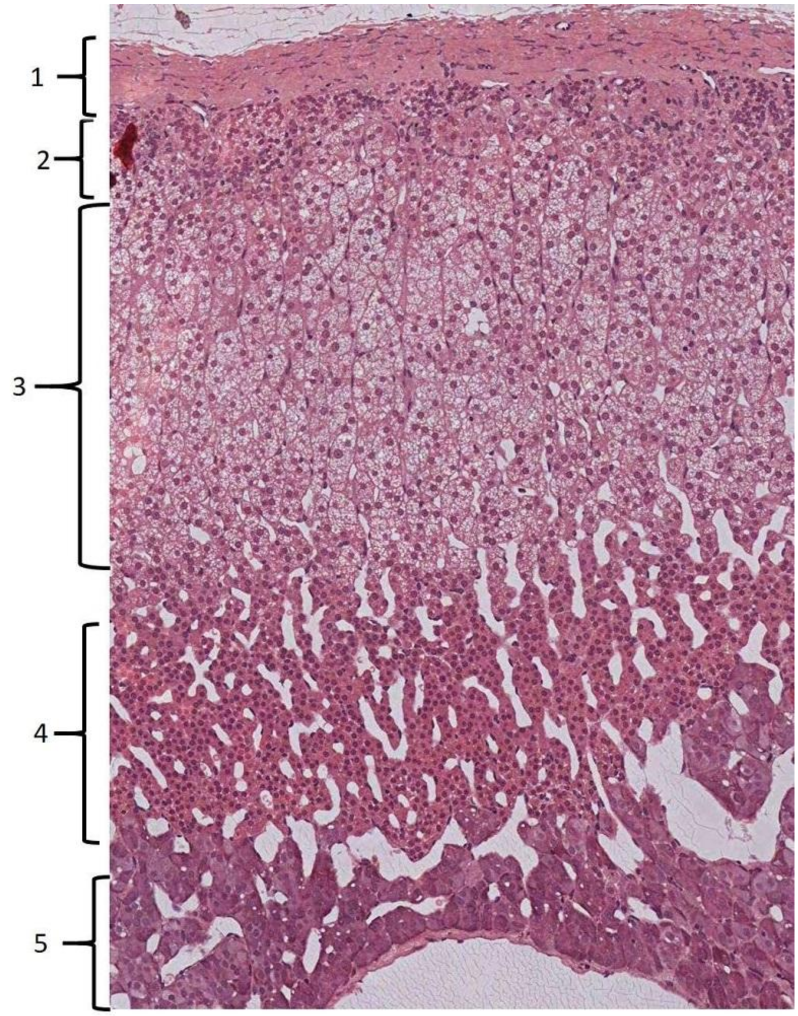
# of Questions: 30

Question #: 1

A 55-year-old man comes to the physician because of a 6-month history of recurrent urinary tract infections and a weight gain of 18 kgs (40 lbs) despite increased physical activities. His blood pressure is 150/100 mmHg and pulse is 70/min. Physical exam shows abdominal striae and an abnormal deposition of adipose tissue in the posterior neck. Laboratory studies show elevated fasting blood sugar of 220 mg/dl (normal 70-100 mg/dl). A CT-scan of his abdomen shows an ovoid mass in the left adrenal gland. Which of the following labelled regions of the adrenal gland in the attached photomicrograph is most likely affected in this case?

A. 1
B. 2
C. 3
D. 4
E. 5

Attachment:



attachment_for_itemid_342589.jpg

Item Description: HCB.1025 - adrenal zona fasciculata

Question #: 2

A 64-year-old man comes to physician because of a 2-months history of loss of sensation on his lower lip and jaw. He underwent a jaw reconstruction surgery 2 months back. Physical examination shows injury to mandibular branch of the trigeminal nerve. Which of the following additional signs and symptoms most likely be seen in this patient?

A. Weakness in clenching the jaw
B. Weakness in closing his upper eyelid
C. Weakness in closing his lower eyelid
D. Weakness in closing his mouth

Item Description: ANAT.0164 Trigeminal nerve mandibular branch (CN V-3)

Question #: 3

A 60-year-old woman comes to the physician because of a 5-day history of passing black, tarry stools. She has a 10-year history of alcohol abuse. Physical examination shows pallor and a distended abdomen. Large dilated veins are visible on her anterior abdominal wall. A stool sample is positive for partially digested blood. Which of the following surgical venous anastomoses is most likely to be performed to relieve her symptoms?

A. Splenic to left renal
B. Inferior mesenteric to superior mesenteric
C. Superior mesenteric to left gastric
D. Left renal to inferior vena cava
E. Splenic to portal

Item Description: ANAT.DG.4465 Portacaval shunts

Question #: 4

A 2-day-old boy is brought to the emergency department by his mother because of a 24-hour history of abdominal distention and vomiting. He has not yet passed his first stool. A barium enema study shows narrowing of a portion of the gastrointestinal tract as shown by the arrow in the image. Which of the following embryological abnormalities is most likely responsible for his condition?

A. Failure of recanalization of the midgut
B. Failure of migration of neural crest cells to the colon
C. Incomplete separation of the cloaca
D. Abnormal rotation of the hindgut
E. Persistence of the proximal part of the omphaloenteric (Vitelline) duct

Attachment:

attachment_for_itemid_242416.jpg

Item Description: ANAT.DG.0414 Megacolon (Hirchsprung's disease).

Question #: 5

A 35-year-old woman is admitted to hospital for prophylactic surgery to remove her ovaries. She has a family history of ovarian cancer. During surgery, which of the following structures is most likely to be ligated to interrupt the main blood supply to the ovary?

A. Transverse cervical
B. Round ligament of uterus
C. Uterosacral ligament
D. Suspensory ligament

E. Proper ovarian ligament

Item Description: ANAT.RP. 0901 Ovarian ligaments

Question #: 6

A 30-year-old woman is brought to the emergency department by her sister because of a 6-hour history of severe lower abdominal and pelvic pain and mild persistent vaginal bleeding. Physical examination shows rebound tenderness in the left lower quadrant of the abdomen. Ultrasonography of the abdomen show fluid accumulation in the peritoneal cavity. Laboratory studies show a positive pregnancy test. A diagnosis of a ruptured ectopic pregnancy is made. Fluid from this rupture will most likely be found in which of the following locations?

A. Vesico-uterine pouch
B. Rectovesical pouch
C. Paracolic gutters
D. Rectouterine pouch
E. Retropubic space

Item Description: ANAT.RP. 0914 Rectouterine pouch

Question #: 7

A 38-year-old woman comes to the physician because of a 6-month history of painful, heavy bleeding during her menstrual period. Her first menstrual period occurred at the age of 15. Ultrasonography of the abdomen shows a cyst on the ovary containing ectopic uterine tissue and the uterus in the normal anteverted position. The angle between which of the following structures best describes this position?

A. Vagina and the cervix
B. Uterine body and cervix
C. Uterine body and bladder
D. Cervix and the rectum
E. Vagina and uterine body

Item Description: ANAT.RP. 0903 Uterine position

Question #: 8

A 15-year-old girl is brought to the emergency room by paramedics because she collapsed during a physical education class at school. Her breath has a fruity odor. Her mother says that she had recently complained of tiredness, dizzy spells, severe thirst and repeated need to urinate. Blood samples are taken. Which of the following laboratory parameters is most likely to be elevated in her blood?

A. Acetoacetate and HbA_2
B. 3-hydroxybutyrate and HbA1c
C. Ammonia and glucose
D. 3-hydroxybutyrate and sorbitol
E. Glucose and bicarbonate ions
F. Glucose and arterial pH

Item Description: Diabetes mellitus - Quiz

Question #: 9

A 27-year-old woman comes to the physician because of a 1-week history of loss of appetite, nausea, weakness, and dark urine. Her temperature is 38ºC (101ºF), pulse is 100/min and respirations are 22/min. Physical examination shows right upper quadrant abdominal tenderness. Laboratory studies of the blood show antibodies against hepatitis B virus. A diagnosis of acute hepatitis B infection is made. This disease often leads to hepatic cirrhosis. During the cirrhotic process there is differentiation of a particular cell type into collagen secreting cells. Which of the following is most likely a normal function of the cells prior to differentiation?

A. Production of bilirubin
B. Phagocytosis of red blood cells
C. Storage of vitamin A as retinyl esters
D. Degradation of exogenous toxins and drugs
E. Synthesis of steroid hormones

F. Lines the biliary tree

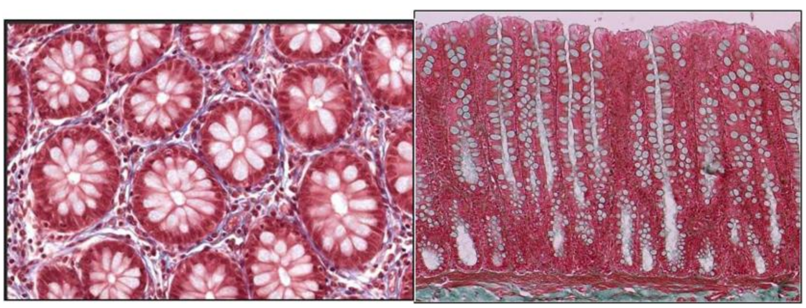
Item Description: HCB.1280 - Ito cells

Question #: 10

A 67-year-old man comes to the physician because of a 2-month history of blood and mucus in his stool, and a 6-month history of dull, aching pain to his lower abdomen. Endoscopy is performed on both the upper and lower gastrointestinal tract. A biopsy is collected for histopathology studies and shown in the attached photomicrographs. Which of the following most likely represents the site of this biopsy specimen?

A. Duodenum
B. Jejunum
C. Ileum
D. Colon
E. Stomach

Attachment:



attachment_for_itemid_342587.jpg

Item Description: HCB.1249 - Regions of the large intestine: colon

---

Question #: 11

A 40-year-old woman comes to the physician because of a 6-month history of frequent headaches and an irregular menstrual cycle. Physical examination shows engorged mammary glands that are producing milk. Laboratory studies of the blood show she is not pregnant. MRI of the head and neck shows an enlarged pituitary gland. Which of the following tumors is most likely the cause of these findings?

A. Lactotrope adenoma
B. Corticotrope adenoma
C. Gonadotrope adenoma
D. Somatotrope adenoma
E. Thyrotrope adenoma

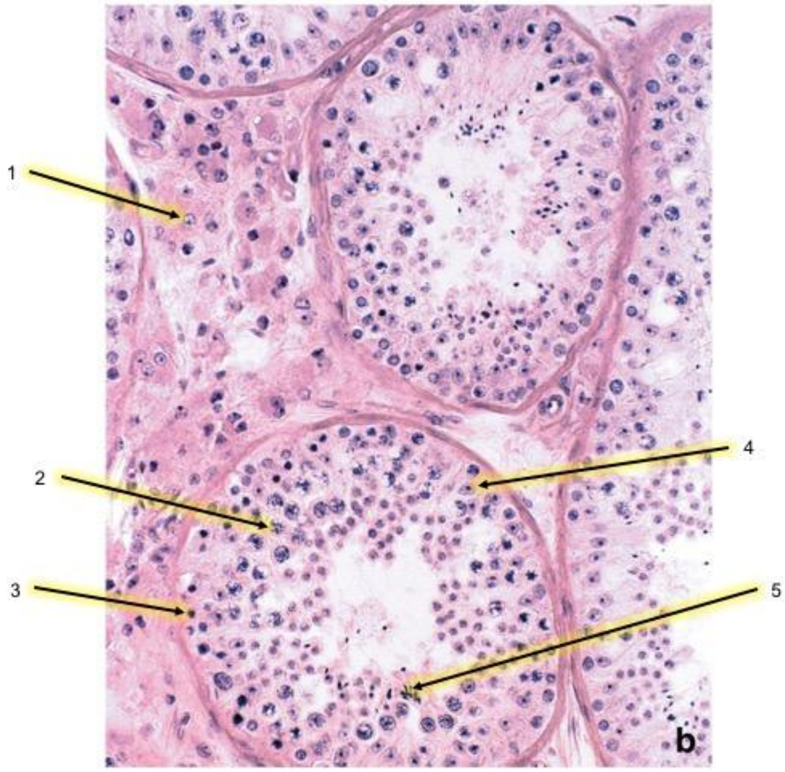
Item Description: HCB.1011 - Anterior pituitary hormones

Question #: 12

A 9-year-old boy is brought to the physician by his mother because of rapid growth and development of pubic hair over the past 6 months. Physical exam shows signs of precocious puberty including development of external genitalia, pubic hair, and facial hair. Palpation and ultrasound of the right testis shows a 3-cm intratesticular mass. Blood tests show elevated testosterone levels. Which of the following cells in the attached photomicrograph most likely gives rise to the testicular neoplasm?

A. 1
B. 2
C. 3
D. 4
E. 5

Attachment:



attachment_for_itemid_342585.jpg

Item Description: HCB.1090 - Leydig cells

Question #: 13

A 25-year-old woman comes to the physician because of an 8-month history of fatigue and shortness of breath on exertion. She says that her menses occur at regular 30-day intervals but the bleeding is heavy and lasts for 6-8 days. Her vital signs are within normal limits. Physical examination shows pale conjunctiva. Laboratory studies show a hemoglobin concentration of 8.2 g/dL (Normal: 12-14 g/dL). She is diagnosed with iron deficiency anemia. Which of the following additional laboratory findings is most likely in this patient?

A. Low arterial P_{CO2}
B. Low O_2 saturation of hemoglobin
C. Increased total O_2 content of blood
D. Normal arterial P_{O2}
E. Low P_{50}

Item Description: PHYS.0261 PaO2 in Anemia

Question #: 14

A 37-year-old woman comes to the physician because of a 6-month history of bone pains. She says that she lives in a place where the sun light is poor. Her vital signs are within normal limits. Physical examination shows mild tenderness of the long bones and the ribcage. Serum studies show a calcium concentration of 7.2 mg/dL (Normal: 8.5-10.5 md/dL) and elevated parathyroid hormone concentration. Which of the following physiologic effects of this hormone is most likely in the kidneys of this patient?

A. Stimulation of Ca^{2+} reabsorption in the distal tubules
B. Increased phosphate reabsorption in the proximal tubules
C. Inhibition of 1 alpha hydroxylase activity
D. Inhibition of 25 hydoxylase activity
E. Activation of PTH2 receptors

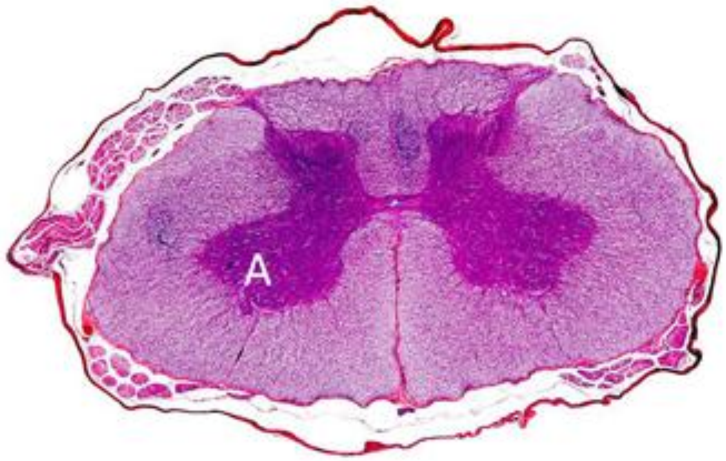
Item Description: PHYS.1058 Ca and PTH

Question #: 15

A 32-year-old woman comes to the physician because of a 1-month history of weakness in her lower limbs. During testing of deep tendon reflexes, cells dwelling in the area indicated by the letter A (see attached figure) are activated by primary afferent neurons arising in the lower limb. Which of the following paired properties best characterize the efferent fibers?

A. Myelinated, pseudounipolar
B. Myelinated, bipolar
C. Myelinated, multipolar
D. Unmyelinated, pseudounipolar
E. Unmyelinated, bipolar
F. Unmyelinated, multipolar

Attachment:



attachment_for_itemid_342322.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0014 Neuronal morphotype

---

Question #: 16

During a post-natal examination, a newborn is observed to have an abnormally small head. Subsequent examinations and investigation findings are consistent with anencephaly. Which of the following is the most likely cause of this condition?

A. Agenesis of the cerebellar vermis
B. Failure of the closure of the anterior neuropore
C. Failure of the closure of the posterior neuropore
D. Cavitation in the spinal canal
E. Anchoring of the spinal cord to the vertebrae

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0547 Structural abnormalities of the developing nervous system

---

Question #: 17

A 14-year-old boy with Down Syndrome is brought to the emergency room because of urinary retention. Physical examination shows flaccid paralysis of both legs, absence of deep tendon and abdominal reflexes. Imaging studies show an abnormality in the anterior half of the spinal cord from Level T5 to T12 consistent with anterior spinal artery infarction. Which of the following is the most likely embryological origin of the affected part of the spinal cord?

A. Roof plate
B. Floor plate
C. Alar plate
D. Basal plate

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0545 Morphogenesis of the spinal cord

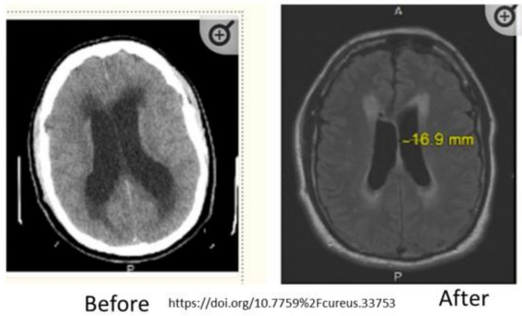
---

Question #: 18

An 86-year-old woman is brought to the physician because of a 4-month history of progressive memory lapses resulting in failure to recognize people. Her gait resembles that of someone with "magnetic feet," and she also experiences frequent urinary incontinence. Following an external ventricular draining (EVD) surgery, the patient's condition improves, except her memory deficit. Brain imaging conducted before and after the EVD surgery reveal a noteworthy change in the ventricular shape. Which of the following conditions is most likely in this patient?

A. Obstructive hydrocephalus
B. Cerebral edema
C. Huntington disease
D. Cerebral stroke leading to paralysis
E. Normal pressure hydrocephalus

Attachment:



attachment_for_itemid_342328.jpg

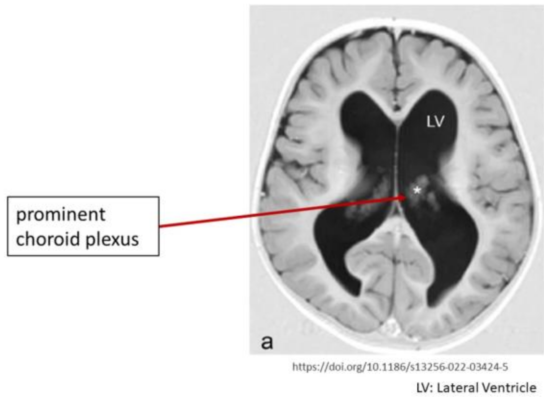
Item Description: Hydrocephalus types

Question #: 19

A 20-month-old girl is brought to the physician by her mother because of a 4-day history of seizures. Physical examination shows accelerated rate of head circumference growth compared to her twin sister and a downward gaze of the eyes. She is unable to walk independently and shows no indications of pain or growth delays. Brain MRI shows enlarged lateral ventricles with a hyperactive choroid plexus (indicated in the image). There is no evidence of cerebrospinal fluid (CSF) flow pathway obstruction. Which of the following terms best applies to the clinical presentation and the radiological findings in this patient?

A. Cerebral vasogenic edema
B. Communicating hydrocephalus
C. Non-communicating hydrocephalus
D. Normal pressure hydrocephalus
E. Hydrocephalus ex vacuo

Attachment:



attachment_for_itemid_342326.jpg

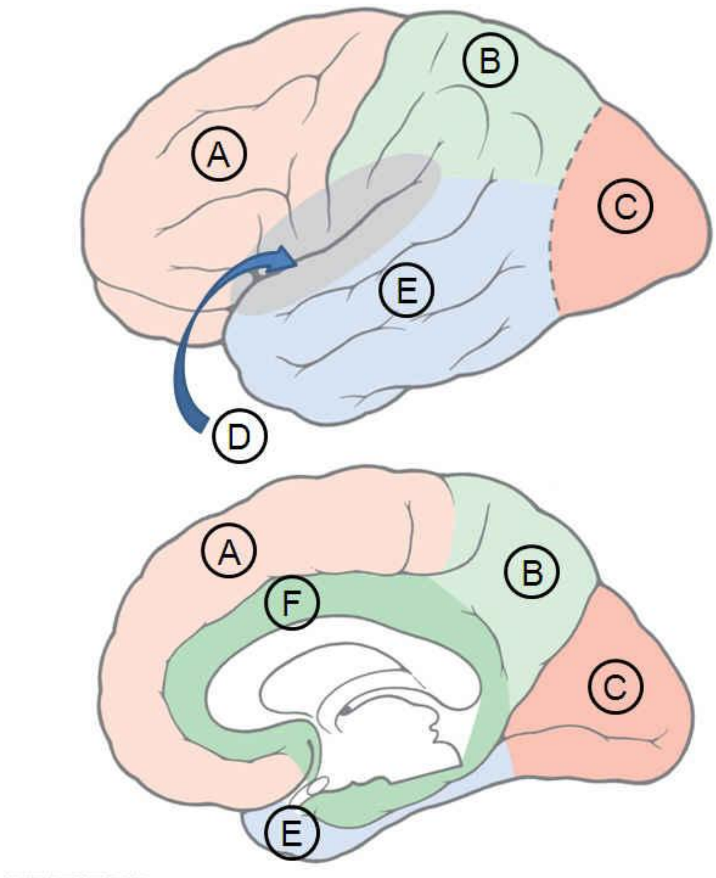
Item Description: Child hydrocephalus

Question #: 20

A 41-year-old man is brought to the emergency department by his daughter because of a 4-hour history of numbness of the left leg. He has a history of hypertension and is a heavy smoker. His Body Mass Index is 35 (obesity). Brain MRI shows a cortical lesion. Which of the following labeled cerebral lobes is most likely lesioned in this patient (see image)?

A. A
B. B
C. C
D. D
E. E
F. F

Attachment:



attachment_for_itemid_342320.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0021 Cerebral lobes

---

Question #: 21

A 62-year-old woman comes to the physician because of a 3-week history of headaches. Neurologic examination shows normal cranial nerve function. Imaging studies show structural malformations (see image). With which of the following structures are the affected vessels most likely associated?

A. Dura
B. Skull
C. Brainstem
D. Spinal cord
E. Cerebrum

Attachment:

attachment_for_itemid_342324.png

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0072 Intracranial subarachnoid vessels

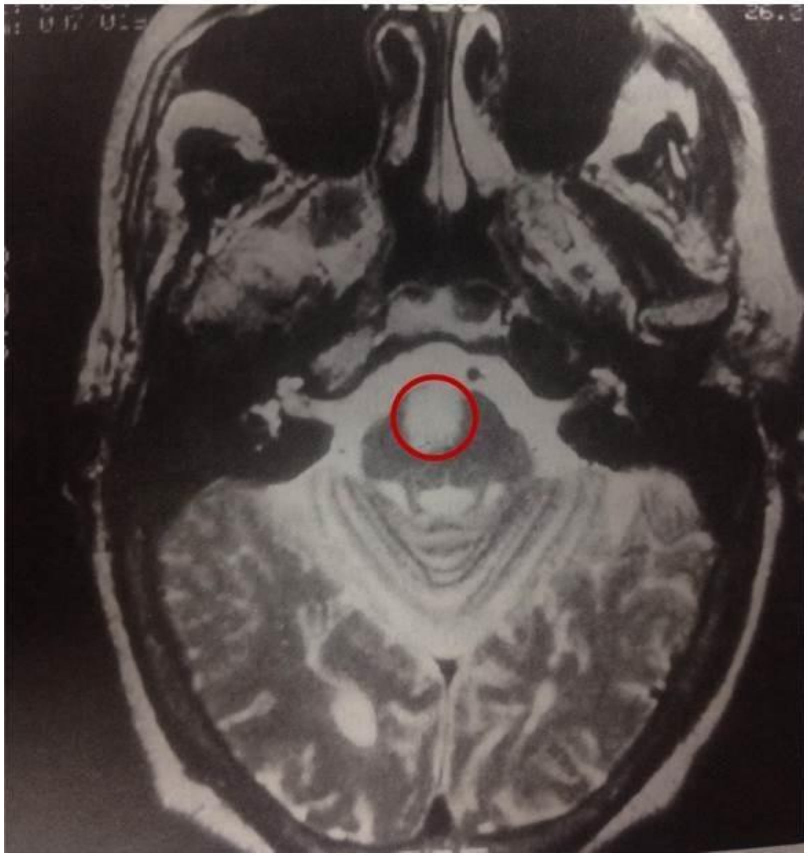
---

Question #: 22

A 73-year-old man is brought to the emergency department because of a 1-hour history of weakness of the limbs and trouble speaking. Imaging studies show a vascular lesion tied to ruptured branches of the basilar artery (see encircled region in the attached image). Which of the following structures is most likely affected?

A. Cerebrum
B. Diencephalon
C. Midbrain
D. Pons
E. Medulla
F. Spinal cord

Attachment:



attachment_for_itemid_342323.jpg

Item Description: SOM.MK.I.BPM2.1.NSCI.0070 Subarachnoid vessels

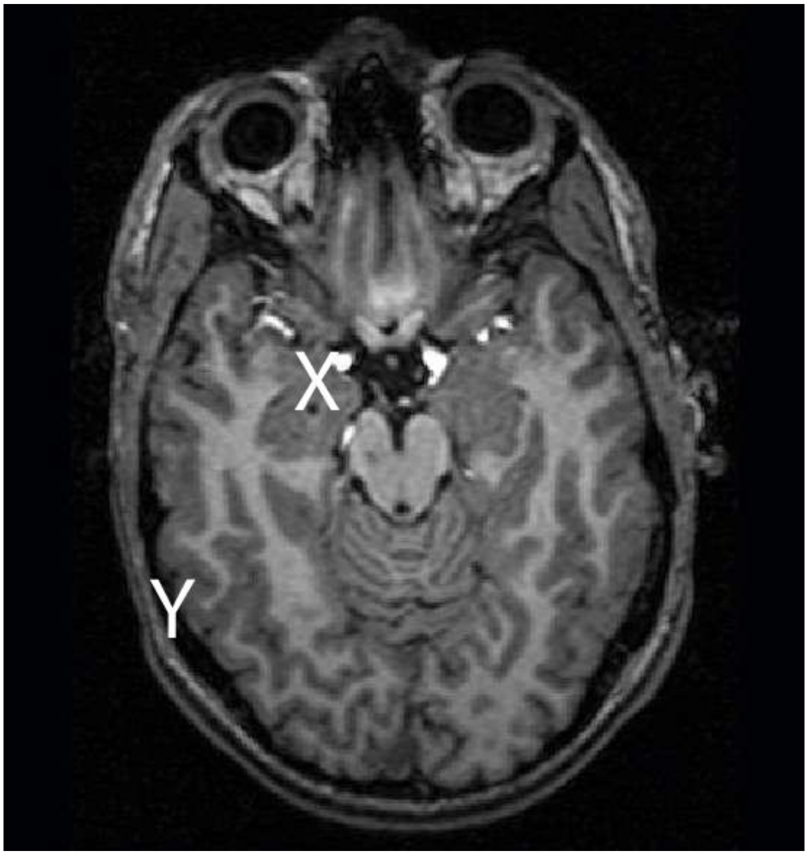
---

Question #: 23

A 52-year-old man is brought to the emergency department because of a 2-hour history of a traumatic brain injury. Imaging studies show a fast-moving fragment of metal entered through the frontal bone of the skull at Site X and lodged at Site Y. Which of the following trajectories is most likely followed by the fragment while passing through the brain?

A. Rostromedial
B. Caudomedial
C. Rostrolateral
D. Dorsolateral
E. Dorsomedial
F. Ventrolateral
G. Caudolateral

Attachment:



attachment_for_itemid_342327.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0009 Anatomical vectors in neuroimaging

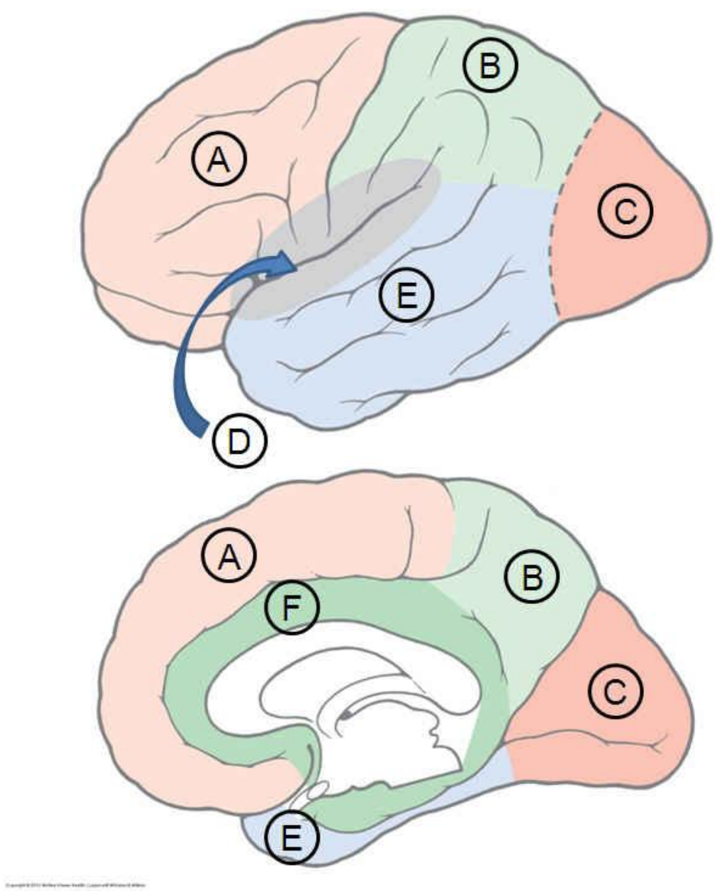
---

Question #: 24

A 59-year-old woman is brought to the emergency department because of a 6-hour history of weakness in her right arm. Physical examination shows diminished strength of the right upper extremity. Which of the following cerebral lobes is most likely lesioned (see image)?

A. A
B. B
C. C
D. D
E. E
F. F

Attachment:



attachment_for_itemid_342316.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0021 Cerebral lobes

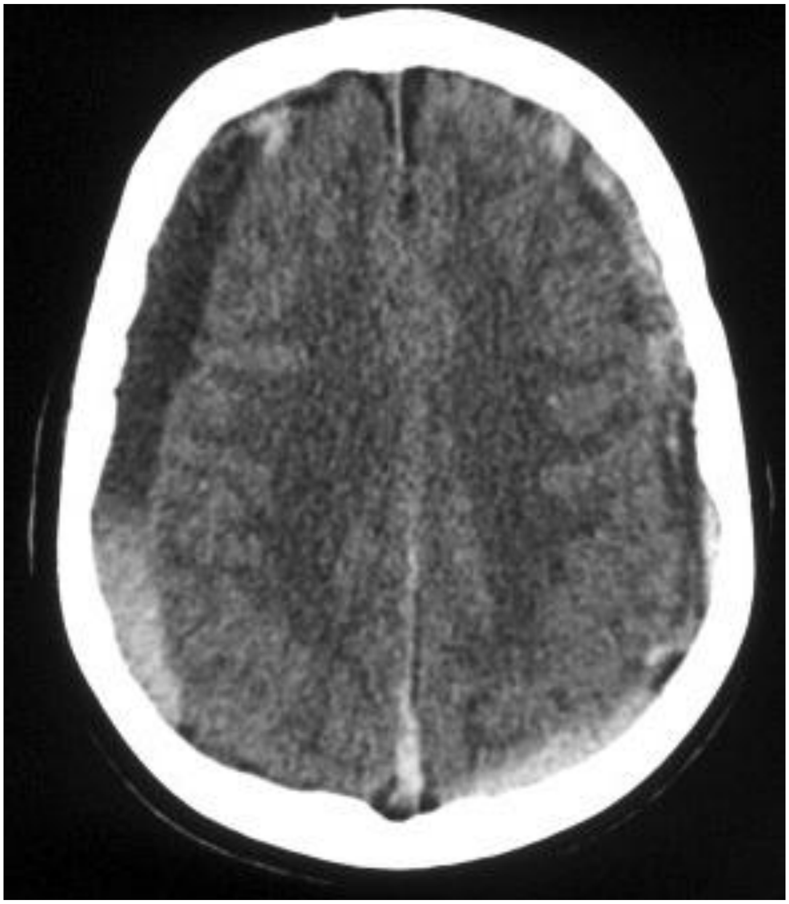
Question #: 25

A 14-year-old boy is brought to the emergency department by his parents because of a 5-hour history of irritability, intense headache, nausea and vomiting that occurred following a blow to the head. Imaging studies show a hematoma (see attached). Which of the following hemorrhagic diagnoses is most likely?

A. Left epidural
B. Left subdural
C. Left subarachnoid
D. Left periventricular
E. Right epidural
F. Right subdural
G. Right subarachnoid

H. Right periventricular

Attachment:



attachment_for_itemid_342319.jpg

Item Description: SOM.MK.I.BPM2.2.NB.1.NSCI.0071 Distinguish hemorrhages

---

Question #: 26

A 23-year-old man comes to the physician because of a 10-day history of progressive weakening of his leg muscles that affect standing and walking. His symptoms started as a tingling sensation in his toes but has since extended symmetrically upward to affect his lower extremities. Physical examination shows reduced knee jerk reflex responses and symmetrical muscle weakness from the hip girdle downward. Cerebrospinal fluid (CSF) analysis shows a significant increase in protein levels and a normal blood cell count and glucose levels. Which of the following cell types is most likely affected in terms of its function?

A. Astrocyte
B. Ependymal cell

C. Microglial cell
D. Oligodendrocyte
E. Schwann cell

Item Description: Glial cell abnormal function

Question #: 27

A 22-year-old woman comes to the physician because of a 2-month history of tingling in her left hand. Neurological examination shows a loss of sensation affecting the left hand. Imaging studies show a small inflammatory lesion on the right side of the brainstem. A central inflammatory demyelinating condition is diagnosed. Which of the following glial cells is most likely affected by the inflammatory process?

A. Microglia
B. Ependymocytes
C. Astrocytes
D. Schwann cells
E. Oligodendrocytes

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Glial cell functions

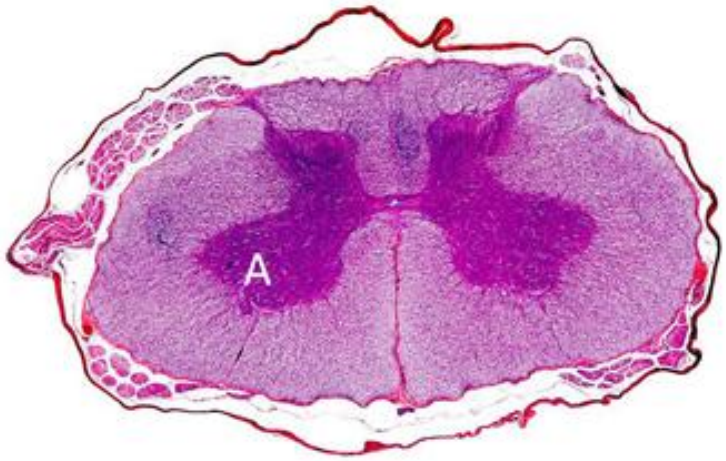
---

Question #: 28

A 24-year-old woman comes to the physician because of a 6-week history of weakness of the lower limbs. Physical examination findings suggest degeneration of cells dwelling in area A (see attached figure). Which of the following properties best applies to the cells occupying the affected area?

A. Upper motor neurons
B. Lower motor neurons
C. Primary somatosensory neurons
D. Commissural fibers
E. Association fibers

Attachment:



attachment_for_itemid_342321.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0014 Neuronal morphotype

---

Question #: 29

A 54-year-old man is brought to the emergency department because of a 3-hour history of nausea and a fainting spell. Imaging studies show hypoperfusion in a small localized area of the brain. An ischemic attack is suspected and adequate treatment is initiated. Four days later, the patient dies from a pulmonary embolism. Brain biopsy shows increased cellularity in the ischemic area with a predominance of microglia. Which of the following functions is most likely associated with the predominant cell in the lesion?

A. Myelin production
B. Removal of debris
C. Scar tissue production
D. Cerebrospinal fluid secretion
E. Cerebrospinal fluid absorption

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Glial cells

Question #: 30

A 39-year-old man comes to the physician because of a 2-month history of intermittent episodes of diarrhea and flushing. Urinalysis shows elevated levels of 5-HIAA. Which of the following enzymes is most likely required for the formation of this metabolite?

A. Glutamate decarboxylase
B. Nitric oxide synthase
C. Catechol-O methyltransferase (COMT)
D. Monoamine oxidase (MAO)
E. Dopamine beta-hydroxylase
F. Arginase

Item Description: HIAA-quiz